HW 4 [Fourier Series]

(a) $\phi(x) = x, \quad 0 \le x \le L$

① Fourier sine series

$$A_n = \frac{2}{L}\int_0^L x \cdot \sin\frac{n\pi x}{L}\,dx = (-1)^{n+1}\frac{2L}{n\pi}$$

$$\phi(x) = x = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{2L}{n\pi}\cdot\sin\frac{n\pi x}{L} = \frac{2L}{\pi}\left(\sin\frac{\pi x}{L} - \frac{1}{2}\sin\frac{2\pi x}{L} + \frac{1}{3}\sin\frac{3\pi x}{L} - \cdots\right)$$

② Fourier cosine series.

$$A_0 = \frac{2}{L}\int_0^L x\,dx = L.$$

$$A_n = \frac{2}{L}\int_0^L x\cdot\cos\frac{n\pi x}{L}\,dx = -\frac{4L}{n^2\pi^2} \text{ for } n \text{ odd}, \text{ and } 0 \text{ for } n \text{ even}.$$

$$\phi(x) = x = \frac{1}{2}A_0 + \sum_{n=1}^{\infty}A_n\cos\frac{n\pi x}{L} = \frac{L}{2} - \frac{4L}{\pi^2}\left(\cos\frac{\pi x}{L} + \frac{1}{9}\cos\frac{3\pi x}{L} + \frac{1}{25}\cos\frac{5\pi x}{L} + \cdots\right)$$

$$= \frac{L}{2} - \frac{4L}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi x}{L}.$$

(b)

① Fourier cosine series of x^2.

$$A_0 = \frac{2}{L}\int_0^L x^2\,dx = \frac{2}{3}L^2$$

$$A_n = \frac{2}{L}\int_0^L x^2\cdot\cos\frac{n\pi x}{L}\,dx = (-1)^n\cdot 2n\pi\cdot\frac{2L^2}{(n\pi)^3} = (-1)^n\frac{4L^2}{n^2\pi^2}$$

$$\phi(x) = x^2 = \frac{1}{2}A_0 + \sum_{n=1}^{\infty}A_n\cdot\cos\frac{n\pi x}{L} = \frac{L^2}{3} + \frac{4L^2}{\pi^2}\sum_{n=1}^{\infty}\left[(-1)^n\cdot\frac{1}{n^2}\cdot\cos\frac{n\pi x}{L}\right]$$

② Set $x = 0$

$$\phi(0) = 0 = \frac{L^2}{3} + \frac{4L^2}{\pi^2}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{1}{n^2}\cdot 1$$

$$\frac{L^2}{3} = \frac{4L^2}{\pi^2}\sum_{n=1}^{\infty}(-1)^{n+1}\cdot\frac{1}{n^2}$$

Therefore

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$